

Harnessing data using symbolic regression methods for discovering novel paradigms in physics

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In recent years, machine-learning methods have profoundly impacted research in the interdisciplinary fields of physics. However, most machine-learning models lack interpretability, and physicists doubt the credibility of their conclusions because they cannot be combined with prior physical knowledge. Therefore, this review focuses on symbolic regression, which is an interpretable machine-learning method. First, the relevant concepts of machine learning are introduced in conjunction with induction. Next, we provide an overview of symbolic regression methods. Subsequently, the recent directions for the application of symbolic regression methods in different subfields of physics are outlined, and an overview of the ways in which the applications of symbolic regression have evolved in the realm of physics is provided. The major aim of this review is to introduce the basic principles of symbolic regression and explain its applications in the field of physics.

machine learning, genetic programming, symbolic regression

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1 Introduction

Machine learning is a branch of artificial intelligence (AI) and computer science capable of providing solutions to problems such as large-dataset analysis and identifying paradigms within quantities of data beyond human-processing capabilities. In recent years, machine learning methods have been used across various research fields and have made significant contributions to physics research. The machine-learning process includes data collection, algorithm selection, feature engineering, model training, and results analysis.

Induction is essential to build a ration-based perception of the world and plays a crucial role in physical research. For example, in the history of physics research, the discovery of the law of gravity marked the origin of classical physics and Planck's fit to blackbody radiation was the doorway into quantum mechanics; both of these were applications of induction. Taking the discovery of the law of gravitation, the induction process can be summarized as follows: observing phenomena (Brahe) → recording data (Brahe) → processing data (Kepler) → inductive analysis (Kepler). A comparison shows that the induction and machine learning processes have some similarities. By thinking of data as a description of phenomena and models as a representation of conclusions, we can say that induction (or machine learning) is the pro-

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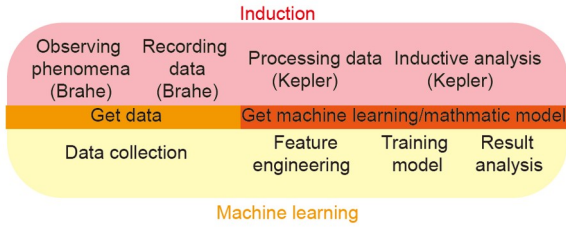


Figure 1 (Color online) The process of discovering the law of gravity is used as an example to juxtapose and compare the general workflow of machine learning and induction.

cess of obtaining a general conclusion (model) from a large number of phenomena (data), as shown in Figure 1.

Machine learning covers several types of algorithms. Different machine learning algorithms have different principles and operation modes that perform well on different problems and return models of different forms. Unfortunately, most major machine learning methods, such as random forests (RF), support vector machines (SVM), and powerful artificial neural networks (ANN), are all black-box models.

Black-box models pay the price of interpretability to ensure model performance because of their complexity. The lack of interpretability raises questions about whether the model is limited in generalizations. Therefore, an abundance of high-quality training data is frequently necessitated to enhance the performance of the model to alleviate overfitting and improve the scope of the application. We believe that a balance exists among the amount of data required for model training, complexity, and interpretability. There is no best machine learning model for all situations, and choosing the most suitable model according to the unique problem is the key to using machine learning methods for research.

Physicists often prioritize interpretable physical logic over obtaining a model that can only provide an answer to a single problem. Therefore, we recommend that physicists use symbolic regression (SR) methods, which are interpretable machine-learning methods that are suitable for small datasets. Sect. 2 of this review provides a basic overview of symbolic regression methods, emphasizing the similarities between SR methods and the data-processing induction step, both of which provide a result that is sufficiently simple for human analysis. In some studies, researchers have used their own SR method to re-implement Newton's workflow. For example, Austel et al. [1] rediscovered Kepler's law and Lemos et al. [2] found an analytical expression equivalent to Newton's law of gravitation.

2 Symbolic regression

2.1 Algorithm overview

The symbolic regression (SR) [3] algorithm is a machine

learning technique that can search the space of mathematical expressions based on a given dataset. During the search, SR identifies optimal mathematical expressions by minimizing the error metrics. Unlike most machine learning algorithms, SR can train a model using even very small training datasets [4], and it outperforms other models for small datasets [5]. In contrast to regression methods built on the known form (such as linear regression [6,7] and kernel-based regression [8]) that can only find the formulas corresponding to the basic form to solve a specific kind of problem, SR searches for the parameters and form of equations simultaneously [9], which provides a more varied searching space and can solve general problems without a priori assumptions [9,10]. This property allows SR to provide interpretable results for physical problems without knowing the equation forms. The free generation of formulas without prior assumptions exponentially increases the complexity of the model, making SR an NP-hard (nondeterministic polynomial-time hard) [11] problem in principle [12]. Considering the limitations of computing power, large instances of such problems cannot be solved using accurate algorithms. In general, researchers adopt effective approximation algorithms to solve these problems and replace the brute-force search with other strategies.

The concept of SR can be traced back to the BACON program [13]. Since then, several strategies have been applied in SR, genetic programming (GP) [14–20], deep learning [21,22], fast function extraction [23], improved programming [24], Bayesian methods [25], MINLP methods [26–29], compressed sensing methods [30–32], etc. [33–38]. Recently, La Cava et al. [39] established a reliable standard to measure the performance of different strategies applied in SR methods.

In the field of physics, GPSR is currently the most widely used SR method with the widest range of applications. In addition to GPSR, the following strategies to achieve SR are also worthy of attention. Symbolic regression based on compressive sensing strategy (CSSR) methods [30,31,40,41] can handle larger feature spaces without additional processing of highly correlated features. This feature makes CSSR methods perform well in mining descriptors of material properties and have been widely applied [32,42–52]. Equation learner (EQL) is based on fully differentiable neural networks, which allow end-to-end training using back-propagation, and implement symbolic regression [53]. Many symbolic regression algorithms have been developed based on EQL in recent years [54–57]. Cozad and Sahinidis [26] proposed the strictly deterministic mixed-integer nonlinear programming (MINLP) [58] formulation for solving the SR problem to global optimality. Optimal solutions have important implications for some physical problems. As a result, MINLP-SR has gained many developments and applications [27,29,59]. Another algorithm worth paying attention to is AI-Feynman [60], which is a specialized SR algorithm for

finding physical formulas. Based on the characteristics of physical formulas, it achieves separation and reduction of features by recursively applying a neural network to simplify the problem so that it can be solved by brute force search.

Although SR methods are diverse, GPSR is the mainstream method, either from the perspective of the development process and performance or from the practical application range. Therefore, we discuss GPSR in detail in sect. 2.2 and most of the applications we introduced in detail in sect. 3 are based on GPSR.

2.2 Genetic programming-based symbolic regression

Inspired by the natural law of “survival of the fittest” and genetics represented by Mendel’s laws, genetic algorithms (GAs) [61,62] were designed. By using genetic algorithms, computational requirements can be significantly reduced compared with exhaustive methods. Koza [14] developed GP, a method that converts mathematical expressions into formula trees. The GP algorithm has been widely used in the computer field, and GPSR is an application of GP in the direction of SR. GPSR relies on randomly generated and “mutated” function expressions to find higher-scoring function expressions. This method reduces computing costs and can obtain satisfactory results in most cases (though it does not guarantee that the final solution will be optimal).

Recently, researchers created a widely used machine-learning library called scikit-learn [63] using Python. During system compilation, developers added GPSR to scikit-learn as an extension in the form of a gplearn module (<https://gplearn.readthedocs.io/en/stable/index.html>). To date, gplearn has become a popular and mature implementation scheme of the SR algorithm in Python, and this method has been widely applied in the fields of physics [64–66]. Various GPSR algorithms have different architectures and implementations. This review will introduce the basic principles of GPSR, using gplearn as an example.

2.2.1 Randomly generating formulas to build a search space

First, researchers define the function set, the terminal set, the algorithm control parameter, the fitness function, and the termination criterion of SR according to the needs of the actual problem. The function set includes mathematical operators, mathematical functions, and conditional operators. The terminal set includes the independent variables and constants that build the equations. The algorithm control parameters include the probability of occurrence of different genetic operations and population size. A fitness function is used to assess the appropriateness of the model. The termination criterion was set as the aborted condition of the loop. Next, the elements in the function and terminal sets are combined to generate unselected random formulas that can

be represented by trees. In these formula trees, the elements in the function are set as internal nodes, and the elements in the terminal function are set as terminal nodes. Each tree corresponds to an independent solution to the problem.

2.2.2 Narrowing the problem space to find a better solution by filtering formulas

After the initial formula is generated, GPSR enters the generation loop. In this step, the genetic SR method evaluates all formulas to select the “fitter” formulas from the current generation formulas. In general, MSE, RMSE, MAE, and other common error metrics used in machine learning can be effectively implemented in general SR problems. Most GPSR methods control the formula complexity by adding a penalty term to the fitness evaluation criterion to avoid overfitting. By adding the penalty term, the formulas that have higher complexity have lower evaluation scores; therefore, they are more likely to be eliminated in the elimination step. Gplearn has a built-in covariant thrift method that considers both the original fitness and complexity of the formula and can dynamically adjust the size of the complexity penalty automatically.

After distinguishing the “fitters” and outputting them to the pool, the algorithm enters the reproduction loop, which performs genetic operations on the “fitters” to generate the next generation. In this study, we use the genetic operations contained in gplearn as an example. A schematic of the genetic operation is shown in Figure 2.

Not limited to the gplearn method, these genetic operations are commonly used in other types of GP. Koza [14] provides a more in-depth understanding of these concepts and their origins. The GPSR loops through the reproduction loop until it reaches the termination criterion set, returns to the beginning of the generation loop, and the generation process is repeated until a generation reaches a preset termination condition. A flowchart of GPSR is shown in Figure 3.

2.3 Related work

Symbolic regression methods have gained increasing attention in recent years. Except for our review, there are numerous other reviews available for readers to reference. Wang et al. [67] mainly explored the application of SR methods in materials science. Angelis et al. [68] provided a comprehensive survey of various SR methods and their applications in diverse domains, such as materials, finance, medicine, engineering, and some non-physical disciplines. Makke and Chawla [69] presented a comprehensive and structured review of SR methods, discussing the advantages and limitations of different SR methods. Liu and Senftle [45] focus on the application of SR (GPSR and CSSR) in the catalysis literature and provide related examples. Distinct from other reviews, our review focused specifically on the

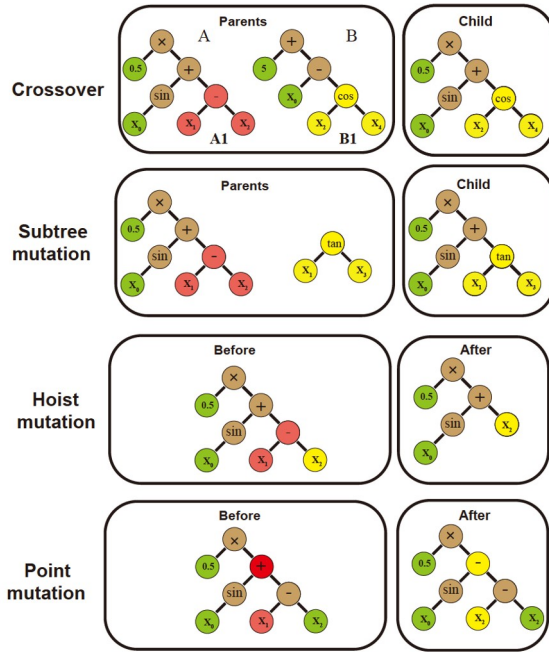


Figure 2 (Color online) Schematic of genetic operations in gplearn. Crossover: Two formula trees are selected from the “fitters”, called formula tree A and formula tree B. From the two formula trees, subtree A1 and subtree B1 are intercepted, subtree A1 is cut and subtree B1 is inserted into A1’s original position. Crossover is the genetic operation which is corresponding to the concept of hybridization in genetics. Subtree Mutation: Cut a subtree of a formula tree and, create a new subtree completely randomly and insert the new subtree into the node where the formula tree was sliced, corresponding to mutation in genetics. Hoist Mutation: Select a subtree of the formula tree, then select part of the subtree, cut the subtree from the formula tree, and graft the part of the subtree to the original position of the subtree. Point Mutation: Point mutation is a more conservative approach than subtree mutation. Point mutation is achieved by randomly replacing a single internal or terminal node within the formula tree accordingly.

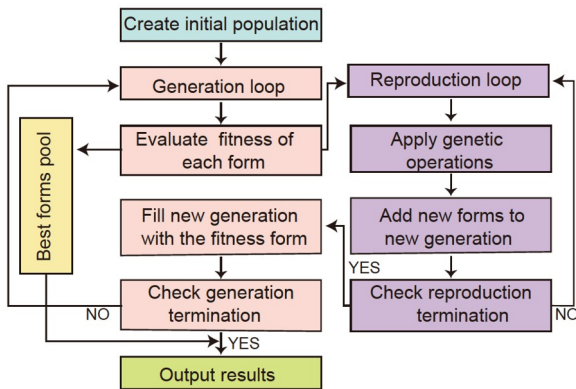


Figure 3 (Color online) Basic flowchart of genetic programming-based symbolic regression.

application of Symbolic Regression methods within several hot topics in physics, and we selected examples that were not only easy to understand but also effectively showcased the practicality of symbolic regression.

3 Application of symbolic regression in the field of physics

As a supervised model, SR can discover hidden mathematical formulas from data to predict target variables. By using SR, potential formulas can be searched from data for physicists to analyze, thereby reducing human workload. Furthermore, SR methods can provide a new cognitive path away from human inertia, leading to conclusions that are easily overlooked or difficult to derive from the human brain. The purpose of this review is to demonstrate the application of SR in physics fields and provide readers with a basic understanding of its application methods and potential future development directions. In each subsection of sect. 3, we present practical examples to demonstrate the application of SR.

3.1 Symbolic regression applied in classical mechanics

Askari and Crevecoeur [70] developed an evolutionary symbolic sparse regression module suitable for multibody system dynamics. They tested their method on a two-DoF spring-mass system, a crank-slider mechanism, and a sliding mass subjected to friction. Liu and Tegmark [71] presented AI Poincaré, a machine learning method that can find conserved quantities from gravitational three-body and other dynamical system trajectory data. They apply their method to five Hamiltonian systems and it discover all exactly conserved quantities. Udrescu and Tegmark [72] built a workflow capable of discovering equations of motion from synthetic video. They used Auto-Encoder as “symbolic progression” to learn trajectories from “video frame” and used SR to discover formulas based on selected features.

In this section, we choose the study of Schmidt and Lipson [9] as a representative example and provide a detailed introduction. This study is considered representative, and the genetic algorithm, Eureqa, designed in this study is regarded as the gold standard for SR and has been widely applied. Schmidt and Lipson [9] used a motion tracking system to record the positions of objects in a single pendulum, double pendulum, and simple harmonic vibration systems over time as datasets and then used partial derivative pairs to create a criterion that can examine the ability of a formula to reflect the intrinsic correlations of a system. Their SR method not only identifies physically significant constant laws but also identifies physically significant equations that are nontrivial conservation laws. They used their method to rediscover the expressions for Hamilton’s equation, Lagrange’s equation, and other geometric and momentum conservation law forms without prior knowledge in the fields of physics, kinematics, or geometry. Depending on the physical quantities provided, the algorithm determines interpretable laws of mechanics or geometry. For instance, in the study of a single pendulum

(Figure 4), the SR method gives the equation of a circle (where c is a constant):

$$(x - 77.72)^2 + (y - 106.48)^2 = c, \quad (1)$$

when given the (x, y) position data over time. This equation describes geometrical circles but does not provide physical insights. When angular velocity ω is added to the given information, the algorithm returns

$$1.37 \cdot \omega^2 + 3.29 \cdot \cos(\theta), \quad (2)$$

which has the form of the Lagrangian equation for the single pendulum:

$$\frac{1}{2}ml^2\omega^2 + mgl\cos(\theta). \quad (3)$$

When acceleration (a) is added to the given information, the algorithm returns

$$2.17a + 0.054\omega - 3.54\sin(\theta), \quad (4)$$

which has the form of a Newton equation for a single pendulum with damping μ along tangential acceleration:

$$\ddot{\theta} + \mu\dot{\theta} + \frac{g}{l}\sin(\theta). \quad (5)$$

3.2 Symbolic regression applied in astrophysics

Matchev et al. [73] used SR on synthetic data for the transit radii of hot Jupiter exoplanets to derive an analytical formula. Wadekar et al. [74] used SR to search for augmentations of the widely used YSZ-M scaling relation. They found that a new mass proxy combined Y200c and gas concentration. In their other work, they modeled the halo environment and used PySR to find expressions that included the effect of the halo environment on its Hi mass [75]. Barsotti et al. [76] used AutoML techniques and the RBEIM framework to predict gravitational waveform surrogates from compact binaries and found that Eureka (GPSR) demonstrated superior performance among 87 methods. Wong and Cranmer [77] demonstrated a strategy to automatically discover interpretable gravitational wave population models and find a new model as proof. Cranmer et al. [78] combined graph neural networks and SR to develop a formula for predicting dark matter density.

We chose Lemos et al.'s study [2] to demonstrate the application of the SR in astrophysics. This study not only demonstrates the ability of SR to obtain comprehensible symbolic formulas by mining astronomical data but also illustrates the application directions of SR in combining physical prior knowledge and integrating it with other machine learning algorithms. This study harmoniously combined prior physical knowledge, SR, and neural networks, demonstrating the potential of neural networks to simplify complex problems and the influential role of prior physical knowledge in guiding machine learning.

This study comprises two steps, as depicted in Figure 5.

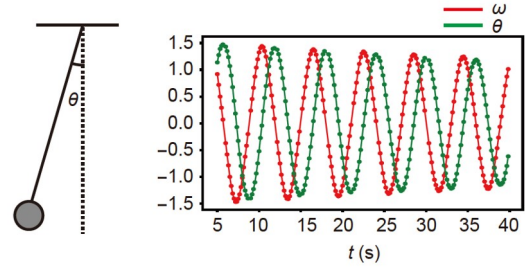


Figure 4 (Color online) Diagrams of the single pendulum and the variations in θ , ω over time (protract based on the supporting material provided by Schmidt and Lipson [9]).

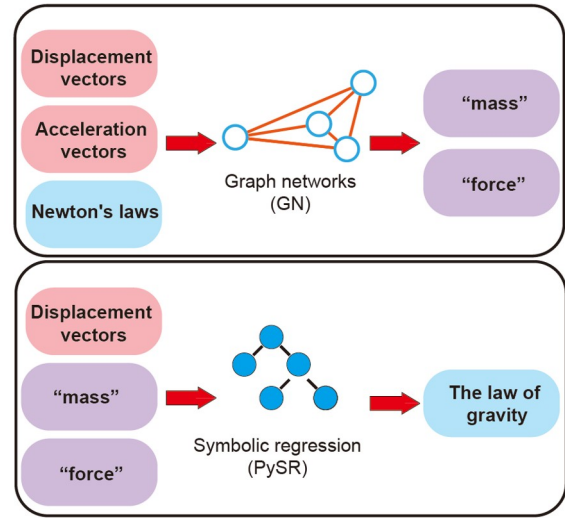


Figure 5 (Color online) Diagrams of the GN step and PySR step. The left side represents the known data, and the right side represents the data fitted by the corresponding machine-learning method. We use a red background to identify the real world observed data, a purple background to identify the “physical quantity” data simulated by the algorithm, and a blue background to identify the physical laws.

(1) Based on the spatial displacement vector between the two corresponding bodies and each body's true observed acceleration vector, using graph networks (GN) [79] to fit the interaction (“force”) vector along each edge and the trainable scalar quantity (“mass”). (2) Based on the displacement vectors, with the “mass” and “force” that were fitted in the previous step, PySR was used to rediscover the law of universal gravitation.

GNs, which are based on “graph” woven from nodes and edges, have a natural advantage in describing interparticle interactions due to their architecture [79]. Based on this advantage, GN-based simulators perform well in predicting the dynamics of physical systems [80,81]. In this work, the authors used a graph network to depict the state of the celestial system, with each celestial object serving as a node and the relationships between celestial objects (relative position and gravity) serving as the edges. This research is based on solar system data from NASA's HORIZONS On-

Line Ephemeris System from January 1980 to January 2013 with a time step of 30 min. The known physical quantities in the training process of the graph network are displacement vector and acceleration, and the “physical quantities” fitted by GN are “mass” and “force”, “force” is given by the edge function of each edge, the edge function consists the “mass” of sender and receiver node, displacement vector trainable parameter ($f_{\text{GN}}(v_{rk}, v_{sk}, e_k; \theta)$). The calculation is reduced by introducing Newton’s third law, such that the forces between a pair of celestial bodies are expressed as positive and negative values of the same value. By introducing Newton’s second law, the “equation” can be constructed as “acceleration” = “net force”/“mass”, and by minimizing the error between the “acceleration” with the acceleration, GN fits “mass” and “force” towards the true mass and gravitation.

PySR is a type of GPSR designed to discover empirical equations for science, and the creators of PySR built a benchmark (EmpiricalBench) to quantitatively evaluate the performance of SR algorithms in scientific applications [20]. In the second step, the authors put the “mass” and “force”, which are fitted by GN, and the displacement vector into PySR to find the gravitational formula. The most accurate formula determined by PySR is

$$\mathbf{F} = -\frac{G_{\text{learned}} M_1 M_2}{r^3} \mathbf{r}, \quad (6)$$

which is equivalent to the universal gravitational law. When the bodies’ orbits are constructed using this formula, they deviate less from the real orbits than the orbits constructed by “force” trained by GN. Furthermore, the authors use the formula found by PySR to replace the edge function in GN and rerun GN to update the value of “mass”. Compared with the “mass” fitted in the first step, the updated “mass” has a lower deviation from the true mass. Finally, the updated “mass” and symbolic formulas were used to redraw bodies’ orbits, which was more in line with the real orbits.

Under the guidance of prior knowledge, this study uses the GN to achieve efficient physical discovery. Subsequently, SR is used to explain the discovery made by GN. The final step replaces the edge function fitted by the GN with the optimal formula derived from SR. Compared with using GN alone, their method, which combines GN and SR, performs more accurately and can provide a more concise and precise description of the physical phenomena. Compared with using SR alone, their method significantly reduced the cost of the equation discovery.

3.3 Symbolic regression applied in fluid mechanics

Computing the exact solutions for complex fluid dynamics is difficult. Consequently, researchers often resort to simpler approximate functions as substitutes for the true function, thereby reducing the problem’s complexity. Manually find-

ing a reliable approximate function is time-consuming; however, as a machine learning method that returns functions, SR has great potential for finding approximate functions. El Hasadi and Padding [82] used semi-supervised GPSR to study the fluid drag experienced by ellipsoidal and spherocylindrical particles with arbitrary aspect ratios and provided new drag correlations. Ansari et al. [83] combined active learning and SISSO to create a framework to obtain symbolic equations between system variables from computational fluid dynamics (CFD) simulations. They proved the effectiveness of their method by recovering an empirical equation for the pressure drop in a bent pipe and finding a new equation for predicting the backflow in a heart valve under aortic insufficiency. Milošević et al. [84] overview existing models that unified the laminar and turbulent formulations of flow friction and used the Eureka mine with two new formulations. Sofos et al. [85] used GPSR to present a Lennard-Jones fluid descriptor based on density and temperature values.

In this section, we present two examples in detail. In the first example, Praks et al. [86] utilized SR to find the explicit approximations of the Colebrook equation. In the second example, Reinbold et al. [87] utilized SR to summarize a model exhibiting a thin-layer fluid with complex spatio-temporal behavior.

The Colebrook equation [88] is widely used for calculating the flow friction factor λ of turbulent flow pipeline transportation problems. However, λ is implicitly given by the Colebrook equation. Therefore, it is worthwhile to find a formula in the form of $\lambda \approx f(Re, \varepsilon/D)$ to process realistic problems. Brkić [89] introduced the existing explicit approximation of the implicit Colebrook equation in their review. The primary objective of this study was to find approximate equations with explicit structures and low computational costs. The SR method, which constrains the model complexity and returns function expressions, is highly adaptive to this research goal.

Praks and Brkić [86] generated 90000 evenly uniformly distributed $\{Re, \varepsilon/D\}$ and calculated their corresponding flow friction factors λ with the Colebrook equation. These data pairs cover turbulent flow in pipes from practically smooth to very rough for Reynolds numbers ranging from 4000 to 10^8 . Initially, they employed Eureka to fit a function that, while not particularly precise, serves as a framework. Subsequently, the fixed-point iteration procedure was used on the imprecise function, and the accuracy continually improved as the number of iterations increased. Finally, the Padé approximation can be used as an inexpensive and accurate approximation of a logarithm.

The best approximation function found by the SR can explicitly represent the flow friction coefficient with only two logarithmic forms, and the relative error between λ calculated by this function and that calculated by the Co-

lebrook equation is as low as 0.13%. Compared with other approximations that can explicitly represent λ [89], this formula can be considered precise. In this study, SR achieved a significant reduction in computational effort with a small loss of accuracy. Figure 6 shows the procedure for searching for mathematical formulas.

Reinbold et al. [87] successfully developed a quantitatively accurate model of a non-equilibrium spatially extended system using high-dimensional, incomplete, and noisy data by combining physical knowledge and SR methods. Prior to this research, there were no reported instances of successfully discovering models based on incomplete data using SR methods. The authors constructed a thin-layer fluidic experimental apparatus that drives fluid flow using the Lorentz force, and the thin-layer fluid flow exhibited complex spatiotemporal behavior. This apparatus was utilized to generate diverse fluid flow states, with the horizontal component of the fluid-layer velocity field recorded as training data (Figure 7 shows a schematic diagram of the experimental setup and workflow).

The machine learning process designed in this study is illustrated as follows. Initially, physical knowledge is used to constrain the model search space and identify the variables and candidate models. Subsequently, a weak formulation of differential equations was used to reduce the effect of noise and dependence on inaccessible features during model training, resulting in a library of candidate models. Finally, the ensemble SR method was applied to select models which balanced accuracy and complexity.

The results show that when the Reynolds number is between 17.8 and 36, the SR method consistently yields the same parsimonious model across different data sets:

$$\partial_t \mathbf{u} = c_1 (\mathbf{u} \cdot \nabla) \mathbf{u} + c_2 \nabla^2 \mathbf{u} + c_3 \mathbf{u} - \rho^{-1} \nabla p + \rho^{-1} \mathbf{f}, \quad (7)$$

where $\mathbf{u}$ is the velocity field, p is pressure, $\mathbf{f}$ is body force, and ρ is fluid density. Here, we give the classic Navier-Stokes equations for readers to compare; the Navier-Stokes equations for incompressible fluids can be written as:

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\rho^{-1} \nabla p + \rho^{-1} \mu \nabla^2 \mathbf{u} + \rho^{-1} \mathbf{f}; \quad \nabla \cdot \mathbf{u} = 0, \quad (8)$$

where μ is the dynamic viscosity. The Navier-Stokes equations for compressible fluids can be written as:

$$\begin{aligned} & \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} \\ &= -\rho^{-1} \nabla p + \rho^{-1} \nabla \cdot \left\{ \mu \left[\nabla \mathbf{u} + \left(\nabla \mathbf{u} \right)^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right] \right\} + \rho^{-1} \mathbf{f}, \end{aligned} \quad (9)$$

where T is temperature, t is time, and $\mathbf{I}$ is the identity tensor. The authors consider that the first term on the right-hand side of the equation describes the advection of momentum, the second term describes the horizontal momentum flux due to viscosity, the third term describes the vertical momentum

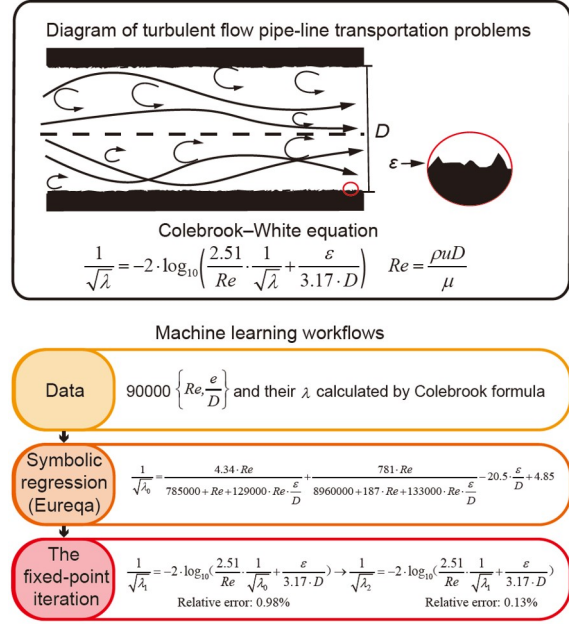


Figure 6 (Color online) Eureqa creates the framework equation, and the relative error of the equation decreases with using one fixed-point iteration increases. D is the hydraulic diameter of the pipe, μ is the dynamic viscosity of the fluid, ρ is the density of the fluid, ε is absolute pipe roughness, u is the mean velocity of the fluid.

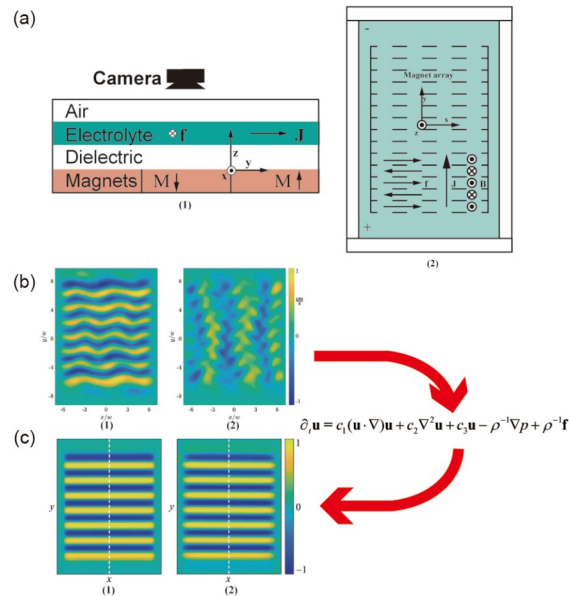


Figure 7 (Color online) (a) shows the rectangular container experimental setup, where ((a)(1)) is the side view and ((a)(2)) is the top view. The dotted lines in ((a)(2)) show an array of 14 permanent magnets and ((a)(1)) show thin electrolyte-dielectric bilayers that are immiscible with each other. When a direct current ($\mathbf{J}$) passes through the electrolyte, the Lorentz force ($\mathbf{f}$) drives the fluid to flow. Panel (b) shows the heatmaps of measured velocity fields in the x - ((b)(1)) and y - ((b)(2)) directions. ((c)(1)) shows the x -component of the depth-averaged Lorentz force computed using experimental measurement of the magnetic field and ((c)(2)) shows the Lorentz force field reconstructed by eq. (1), wherein the Reynolds number corresponding to (b) and (c) is 22.17. (a) is reproduced from Figure 1 in the article by Reinbold et al. [87], and (b) and (c) are original figures sourced from Figure 1 in the article by Reinbold et al. [87].

flux due to viscosity, and the fourth and fifth terms describe (isotropic) internal stresses and external stresses, respectively. In general, the overall structure of the model determined by SR approximates the Navier-Stokes equation, an important equation in the field of fluid mechanics research.

A noteworthy aspect of this study is that the model constructed using SR can almost perfectly reconstruct the external force field driving the fluid flow, whereas the force field data are not included in the training dataset. This demonstrates that the introduction of prior knowledge not only improves the efficiency and success rate of SR methods but also extends the input data, allowing researchers to use incomplete data to train models that can fully describe the physical problem. Furthermore, based on the generated model, researchers can reconstruct information not contained in the training set.

In the first example, we demonstrated the ability of SR to mine simple formulas from datasets to reduce the computational cost of solving physical problems. This example shows that relying on the high computing power of computers, the SR method can quickly and automatically discover approximate equations with varying degrees of precision and complexity for researchers to choose from. In the second example, we focused on how SR can discover complete physical models based on incomplete experimental data by introducing domain knowledge from physics. This example shows that SR, as a white-box model, can be closely integrated with prior knowledge and, therefore, has the unique advantage that black-box machine learning models do not possess.

3.4 Symbolic regression applied in molecular dynamics

The interatomic potential is fundamental to molecular dynamics research, directly related to the success of calculations and the reliability of the conclusions. The calculation of the potential energy surface with first-principles methods can be computationally expensive. To solve this problem, researchers have used machine learning techniques to enhance the efficacy [90].

In the molecular dynamics field, symbolic regression is typically implemented using GP and its variants. Makarov and Metiu [91] created a GPSR method named directed genetic search and used it to rewrite the Morse potential as an example, demonstrating that the method can fit the potential of a diatomic or a triatomic molecule. Brown et al. [92] used a new parallel GP approach that employed large-scale computations to rediscover the exact potential functional forms and parameter values. In their study, the Hierarchical fair competition (HFC) strategy performed better than the standard generational evolutionary strategy. Bellucci and Coker [93] used a parallel multilevel GP to obtain an accurate empirical valence bond model for proton transfer in 3-

hydroxy-gamma-pyrone in the gas phase and protic solvents. Hernandez et al. [94] discovered a multibody potential model for copper by using GPSR and comprehensively evaluated their model. Their SR potential function has a speed similar to that of the EAM-type models and requires less training data than neural network potentials. In their recent study, they test the transferability of their Cu interatomic potential models in other face-centered cubic transition metals. To the elements close to Cu on the periodic table, their model performs well in balance of accuracy and complexity [95]. For various states of Lennard-Jones (LJ) fluids, Papastamatiou et al. [96] used SR to discover analytical expressions that describe the self-diffusion coefficient as a function of other physical properties. In this study, the application of methods in data subcategories can provide better equations for the description of the data.

This review provides a detailed introduction to the study of Slepoy et al. [97] to briefly demonstrate the basic application of SR in the field of molecular dynamics and showcase the ability of GPSR to mine formulas from datasets. Burlacu et al. [98] discussed the application of SR models of interatomic potentials and provided a method that can model atomic potentials from snapshots of atomic positions and associated potential energy. Slepoy et al. [97] combined GPSR and parallel tempering (PT) [99] to build the SR method. PT combines local and global sampling, and its introduction can prevent GPSR from falling into local optima. They examined whether their methods could discover compact force-field functions that accurately represent the potential energy surface of condensed systems. Therefore, the Lennard-Jones potential function, a simple, classic, and widely used function in the field of molecular modeling and condensed matter systems, was chosen for testing. The Lennard-Jones potential is a two-body potential used to simulate the potential energy of interactions between two non-bonding atoms or molecules (such as rare gas elements). Details of the Lennard-Jones potential is shown in Figure 8.

The training set consists of 10 configurations of 10 particles placed in a three-dimensional domain in the field of $3\sigma^*$ $3\sigma^*$ $3\sigma^*$ and the authors defined the particle distribution condition and effective pair distances in the range. By generating atomic configurations and calculating the energy of each configuration using the Lennard-Jones potential, four independent training sets were constructed. As shown in Figure 9, the SR method found algebraically equivalent functions to the Lennard-Jones potential function in three of them: functions (1)-(3). Function (4) can also be considered as a usable approximate function of the Lennard-Jones potential, provided it is in the commonly used interval. The relative error was $<10\%$.

We consider that SR holds the potential to positively impact the fitting of potential energy surfaces by discovering symbolic expression formulas. These formulas may be

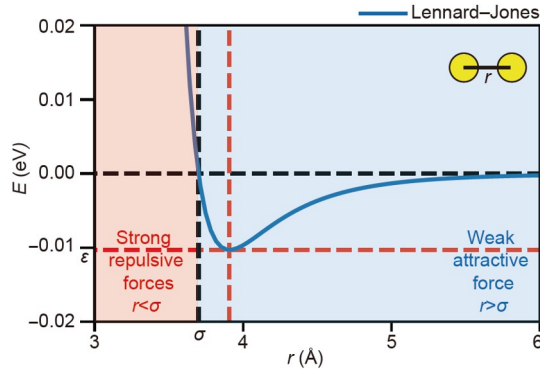


Figure 8 (Color online) Diagrams of Lennard-Jones potential. The curve is plotted based on the change in Lennard-Jones potential energy between argon and argon. Readers of this article should be aware that Lennard-Jones potential is a semi-empirical model. ϵ is the depth of the potential well, reflecting the strength of the attraction between two atoms. σ is the distance between atoms when the Lennard-Jones potential is equal to 0.

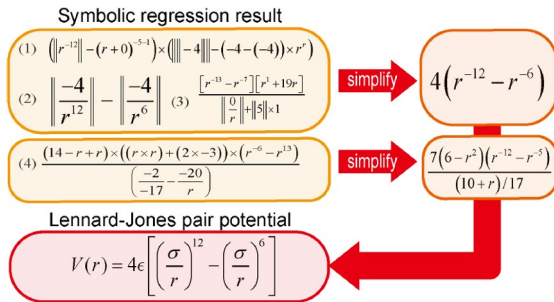


Figure 9 (Color online) Lennard-Jones potential and the equations searched by symbolic regression result (functions (1)-(4)).

helpful in reducing the computational requirements for fitting potential energy surfaces or enhancing the accuracy of the fitted potential energy surfaces.

4 Conclusions

In this review, we delve into the parallels between machine learning and induction pipelines, elucidating how symbolic regression methods serve as useful tools in physics research. To provide a fundamental understanding of symbolic regression, this review introduces the principles and characteristics of genetic programming-based symbolic regression. Sect. 3 of the review navigates through the practical applications of SR in various physics subdivisions. Based on existing research, we examine the combination of SR methods with physics and deliberate on potential avenues for future research.

SR provides advantages for small datasets, offering potentially interpretable results in the form of symbolic expressions. Prior to applying SR, researchers need to ensure the alignment between the characteristics of the problem and

the fundamental attributes of the SR method. Given the diverse implementation methods of SR, it becomes imperative to understand their distinct characteristics.

The combinations of SR methods and physical knowledge can be multifaceted. Researchers have the option to employ existing SR methods for model training and leverage prior knowledge in various stages of machine learning as a guide. Another approach involves crafting specific SR methods that integrate prior physical knowledge. The incorporation of prior knowledge often yields positive outcomes, as elucidated in sect. 3. Not only can it impose constraints to drive conclusions with enhanced physical meaning, but it can also serve to augment training data and mitigate the impact of data noise and missing data.

This review covers the subfields of classical physics that which analytical formulas are dominant. SR is also noted to be applied in other subfields, such as nuclear physics [100], and condensed matter physics [65]. Since SR can be considered as an interpretable machine learning approach, it naturally can be applied to black-box problems. For example, the exchange-correlation function is a long-term puzzle in the field of density functional theory. Recently, Ma et al. [101] created a symbolic functional evolutionary search (SyFES), providing interpretable accurate exchange-correlation functionals in the field of quantum physics. We anticipate that the development of new SR methods based on the needs of the specific topics would definitely expand the application scope of SR in other physics sub-fields.

In summary, symbolic regression is a machine-learning method that has many similarities with induction. It can mine symbolic formulas from extensive data. Depending on the requirement, researchers have the option to either try to provide a physics-based interpretation of the obtained formulas or simply use them as a mathematically convenient alternative to solve computationally expensive problems. We consider that SR holds vast potential for further development and application within the field of physics. However, in scientific discovery, scientists usually encounter high-dimensional inputs (e.g., images, observations) for which they do not know what collective/macroscopic variables (e.g., entropy, energy, quasi-particles) are important. Thus, combining SR with the discovery of these important variables should be a very important direction.

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